SCS Flood Hydrograph Software: Complete User Guide

Software Overview

SCS Flood Hydrograph is a specialized, user-friendly application developed in Python with a graphical user interface (GUI). Using input data that include a rainfall hyetograph, a Curve Number (CN) map, a flow-length map, and physical watershed characteristics, the software calculates the flood hydrograph at the watershed outlet and presents the results as a table and graph.

The equations and computational methods used in the software follow established technical references, including NRCS Technical Release 55 (TR-55) and the HEC-HMS Technical Reference Manual. The software supports both the lumped SCS method and the distributed ModClark method. It is intended to provide an accessible and efficient tool for watershed and water-resources engineers, researchers, and students.

Competitive Advantages

SCS Flood Hydrograph was designed to address limitations encountered with conventional workflows and comparable software. Its main advantages are:

- Distributed ModClark method with direct raster input: similar in principle to the capability available in HECHMS, but with a simpler user interface and direct loading of CN, flow-length, and rainfall rasters without complex preprocessing.
- Elimination of intermediate data conversion: raster data can be used directly, avoiding conversion to intermediate formats such as Excel or CSV and reducing manual effort and potential human error.
- Open-source code: transparent code allows researchers to inspect and verify equations and algorithms and supports further development and customization.
- Fast execution: array-based computation with NumPy and memory optimization can substantially accelerate processing.
- Free, portable executable: the software is distributed as a standalone EXE file and can run on Windows without a separate Python installation, additional library installation, or administrator privileges.

The computational methods implemented in the software are described below.

1. The SCS (Soil Conservation Service) Method

The SCS method, now associated with the Natural Resources Conservation Service (NRCS), was developed in the 1950s to estimate rainfall-runoff from small and medium-sized watersheds. It separates the rainfall-to-flood process into two distinct components:

- Loss: estimates the portion of rainfall lost to infiltration, depression storage, and evaporation, and the portion remaining as runoff (excess rainfall). Losses are estimated using the Curve Number (CN) method.
- Transform: routes the excess rainfall toward the watershed outlet to produce a flood hydrograph (discharge versus time). The software provides two transformation approaches: the lumped SCS unit hydrograph and the distributed ModClark method.

1.1. Initial Abstraction and Excess-Rainfall Calculation

The SCS method assumes that runoff does not occur until rainfall exceeds an initial threshold called initial abstraction (Ia). According to NRCS technical guidance (TR-55, 1986), potential maximum retention (S) and cumulative excess rainfall (Pe) are calculated as follows:

$$S = (25400 / CN) - 254 \text{ (mm)} \quad (1)$$

$$Ia = \lambda S \text{ (the default value of } \lambda \text{ is 0.20; the user can change it)} \quad (2)$$

$$Pe = (P - Ia)^2 / (P - Ia + S), \text{ when } P > Ia; \text{ otherwise } Pe = 0 \quad (3)$$

Potential maximum retention (S) represents the soil's potential water-retention capacity and is inversely related to CN. A higher CN, as typically associated with less infiltrative surfaces, produces a lower S value.

Initial abstraction (Ia) is commonly set to 20% of S, although this coefficient can be modified by the user.

Cumulative excess rainfall (Pe) is the portion of cumulative rainfall that becomes runoff after initial abstraction and continuing retention are accounted for.

Where:

- CN = Curve Number (1–100)
- S = potential maximum retention after runoff begins (mm)
- P = cumulative rainfall depth (mm)
- Ia = initial abstraction (mm)
- Pe = cumulative excess rainfall (mm)

Because rainfall occurs in time steps, the software first calculates cumulative excess rainfall and then differences successive values to obtain incremental excess rainfall for each time step. These increments are used in the transformation component.

2. Lumped SCS Method: Dimensionless Unit Hydrograph

In this method, the entire watershed is treated as a single unit. Spatial differences, such as variation in vegetation cover or slope, are not represented explicitly. The software uses the NRCS dimensionless unit hydrograph (2004 guidance) to transform excess rainfall into runoff.

2.1. Timing Parameters

The timing of the flood response must be specified to construct the hydrograph:

- Lag time (Tlag): the time interval between the centroid of excess rainfall and the peak of the hydrograph.
- Time to peak (Tp): the time from the beginning of the unit-hydrograph response to its peak.
- Base time (Tb): the total duration of the unit hydrograph. Its default value is 2.67 times Tp, but the user can enter a different value.

$$Tlag = 0.6\ Tc \quad (4)$$

$$Tp = \Delta t / 2 + Tlag \quad (5)$$

$$Tb = 2.67\ Tp \text{ (default)} \quad (6)$$

- Tc = time of concentration (hours), entered by the user
- Δt = time step (hours)
- Tp = time to peak of the unit hydrograph
- Tb = base time of the unit hydrograph

2.2. Peak Discharge of the Unit Hydrograph

For 1 mm of excess rainfall, the peak discharge of the unit hydrograph (qp) is calculated qs:

$$qp = 0.208\ A\ /\ Tp \qquad (7)$$

Here, A is watershed area in km² and qp is in m³/s. The coefficient 0.208 reflects unit conversion and the SCS hydrograph shape factor of 0.75.

2.3. Dimensionless Unit Hydrograph

Instead of evaluating a complex mathematical expression, the method uses a standard table relating the discharge ratio (q/qp) to the time ratio (t/Tp). The software stores the table as an array and uses linear interpolation to estimate discharge at intermediate times.

$$\frac{qt}{qp} - f\left(\frac{t}{Tp}\right) \qquad (8)$$

2.4. Convolution

Finally, the software multiplies the excess rainfall generated at each time step by the corresponding unit-hydrograph ordinates and sums the resulting hydrographs through convolution to obtain the final flood hydrograph:

$$Q(i) = \Sigma\ Pe,j \times UH(i - j + 1) \qquad (9)$$

Pe,j = incremental excess rainfall in time step j (mm); UH = unit hydrograph (m³/s per unit depth, as used in the implementation).

3. Distributed ModClark Method

The ModClark method is designed for large or spatially heterogeneous watersheds. The watershed is divided into raster cells, and calculations are performed for each cell using its local characteristics, including CN and distance to the outlet. The method was developed for spatially distributed runoff modeling and is implemented in HEC-HMS (Kull & Feldman, 1998).

In ModClark, travel time to the outlet is calculated for each raster cell. Cells are then grouped by travel time to construct a time-area histogram, which forms the basis for transforming excess rainfall into an inflow hydrograph. The resulting hydrograph is routed through a linear reservoir with storage coefficient R to represent watershed storage effects.

3.1. Travel Time

Water falling on each cell requires time to reach the watershed outlet. Under the assumed linear relationship, travel time is proportional to the distance from the cell to the outlet.

$$Tcell = (Dcell / Dmax) \times Tc \qquad (10)$$

Dcell = flow length from the cell to the outlet (m), read from the flow-length raster
Dmax = maximum flow length in the watershed (m)

3.2. Distributed Excess-Rainfall Calculation

For each cell, the software uses the cell-specific CN value from the CN raster and the rainfall for each time step. If a rainfall raster is available, rainfall is spatially scaled at the cell level. The excess rainfall for each cell is then calculated using the SCS excess-rainfall equation.

3.3. Generation of the Inflow Hydrograph to the Linear Reservoir

After excess rainfall has been calculated, the contribution from each cell is computed for every time step.

$$\Delta Qcell\ (k) = Pe,cell \times Acell \times 1000 / (\Delta t \times 3600) \quad (11)$$

Acell is cell area (km²), Pe,cell is incremental excess rainfall for that cell and time step (mm), and the factors 1000 and 3600 are used for unit conversion to m³/s. The cell contributions are shifted according to their travel times (Tcell) and accumulated to form the inflow hydrograph:

$$Qin\ (k) = \sum\sum \Delta Q_{cell}\ (k') \quad ; \quad k' = \text{rain}_{\text{step}} + \text{delay_step} \quad (12)$$

$$\text{delay_step} = \lfloor \text{Tcell}/\Delta t \rfloor, \text{ min} = 1 \quad (13)$$

3.4. Linear Reservoir Routing

After surface runoff moves across the watershed, some water is temporarily stored and released gradually. To represent this effect, the software routes the inflow hydrograph through a linear reservoir with storage coefficient R (hours). This reduces the flood peak and extends the hydrograph duration, similar to a low-pass filter:

$$C = \Delta t / (R + \Delta t / 2) \quad (14)$$

$$Qout\ (i) = C.\left(Qin\ (i) + Qin\ (i-1)\right) + (1 - 2C).Qout\ (i-1) \quad (15)$$

- R = storage coefficient (hours; user input)
- Qin = inflow hydrograph to the reservoir (m³/s)
- Qout = final outflow hydrograph (m³/s)

This equation is derived using the bilinear transform and is also used in HEC-HMS (USACE, 2000).

4. Comparison of the Lumped and Distributed (ModClark) Methods

Selection of the appropriate method depends on the main differences summarized in Table 1.

Feature	Lumped method	Distributed method
Data basis	Watershed-average values (average CN, Tc, total rainfall)	Spatial rasters (CN, flow length, and optional distributed rainfall)
Spatial variability	No; the watershed is treated as one unit	Yes; each raster cell is calculated separately
Transformation mechanism	SCS dimensionless unit hydrograph	Time-area histogram plus linear reservoir
Additional parameter	None	Storage coefficient R (hours)
Raster data requirements	CN raster optional if average CN is entered; watershed area required	CN and flow-length rasters required; rainfall-depth raster optional
Execution speed	Very fast	Moderate; depends on raster dimensions
Best suited to	Small, relatively homogeneous watersheds or limited data	Large, heterogeneous watersheds with adequate spatial data

Advantages and Limitations

- Lumped method: advantages include simplicity, low data requirements, and fast execution. Its limitation is that it ignores spatial heterogeneity, which can produce unrealistic estimates for large watersheds.
- Distributed method: it represents spatial variation in CN, flow length, and rainfall and may provide better estimates of peak discharge and hydrograph duration. It requires good-quality raster data, takes longer to run, and is sensitive to errors in the flow-length raster.

5. Input Data and File Formats

5.1. Text Input

The only required text input is the rainfall hyetograph, supplied as a .txt file. In the example shown in Figure 1 of the original guide, the first column contains time (hours) and the second column contains cumulative rainfall for the selected storm event or return period (mm).

```
0.25    25.2
0.5     33.0
0.75    37.9
1       41.5
1.25    44.4
1.5     46.9
1.75    49.0
2       50.9
2.25    54.9
2.5     57.1
```

Figure 1 - Contents of the txt file related to the precipitation hytograph. From the left, the first column is the time of the precipitation hytograph (in hours) and the second column is the cumulative precipitation of the event or return period (in mm).

5.2. Raster Inputs

The raster inputs required for the distributed method are the CN raster, rainfall-depth raster (for the selected event or return period), and flow-length raster (m). The software supports GeoTIFF (.tif) and ASCII grid (.asc) formats. GeoTIFF is strongly recommended because it preserves spatial metadata, coordinate reference information, and the definition of NoData values.

All rasters must use a projected coordinate reference system, such as UTM. A geographic coordinate system such as WGS84, whose units are degrees, can lead to incorrect pixel-area calculations and therefore erroneous runoff volume and peak-discharge estimates.

The three rasters must have exactly the same coordinate system, cell size, number of rows and columns, and spatial extent so that cells overlap correctly. Before loading them, verify that their extent and geotransform are aligned in a ArcGIS application.

NoData values must be correctly defined in all rasters so that cells outside the watershed or without valid data are excluded from calculations. The software reads NoData from raster metadata and excludes matching cells and NaN values from calculations, including average CN, maximum flow length, and cell contributions to the

hydrograph. A value such as −9999 is recommended for NoData; zero should not be used because some GIS applications use zero as a default NoData value, which can cause valid values—especially in a flow-length raster—to be excluded.

If a distributed rainfall raster is not available, the software uses uniform watershed rainfall from the text file, and no rainfall raster is needed. If a rainfall raster is supplied, its values must be in millimetres and consistent with the total storm rainfall in the text file so that spatial scaling is applied correctly.

6. Installation and Execution

SCS Flood Hydrograph is distributed as a standalone, portable executable and does not require installation, a separate Python interpreter, additional libraries, or administrator privileges. It has been tested and is recommended for Windows 10 and later. Windows 8.1 may technically run the executable because the bundled Python interpreter supports it, but use is not recommended because Microsoft ended official support for Windows 8.1 in January 2023 and the software has not been sufficiently tested on that system. Windows 8.0 is not supported by the bundled Python interpreter; users should upgrade to Windows 10 or later.
The minimum memory requirement is 4 GB RAM (8 GB recommended), with at least 300 MB of free disk space for temporary extraction of supporting files.

To run the software, copy the EXE file to a suitable folder and double-click it. Extract the executable from any ZIP archive before running it; launching it directly from inside a ZIP file can cause errors when temporary files are loaded. Avoid very long paths and folder names, particularly on older operating systems. On first launch, startup may take 5–15 seconds; subsequent launches typically take less than 5 seconds.

Because the executable is packaged with PyInstaller, Windows SmartScreen may display “Windows protected your PC” on first launch. If you trust the file source, select “More info” and then “Run anyway.” Some security products, particularly Kaspersky and Norton, may incorrectly flag PyInstaller-packaged executables. If this occurs, add the file to the antivirus exclusions list only after verifying that the file is from a trusted source.

After a successful first launch, the software activates its local license state and provides a 10-day trial. The remaining trial period is displayed in the status bar at the bottom of the main window.

Troubleshooting

If the application does not open, closes immediately, or displays no visible error, try the following checks:

- Ensure that the executable has been fully extracted and that extraction completed without errors.
- Check whether Microsoft Visual C++ Redistributable (2015–2022, usually x64) is installed. The GDAL library used to read rasters depends on these components. They are commonly present on Windows 10 and 11, but manual installation may be necessary on Windows 8.1.
- If antivirus software blocks execution, verify the file source and add it to the exclusions list if appropriate.
- Ensure that drive C: has at least 300 MB of free space for temporary files extracted by PyInstaller to the Windows TEMP folder.
- If the problem persists, place the executable in a folder with normal write permissions, such as My Documents, and avoid protected locations such as Program Files, where administrator permissions may be required.

7. Software License and Activation

SCS Flood Hydrograph is described as free and open-source software; however, use of the executable requires an activation license to support continued development. After the first launch, each user receives a free 10-day trial

during which all program features are available without restriction. The remaining trial time is continuously displayed in the status bar at the bottom of the main window.

After the trial expires, the software locks automatically. Continued use requires an Activation Key from the developer. A key can be requested by emailing bahram368@gmail.com. Each key is issued as a 10-character alphanumeric string and is valid for 10 days by default. Longer validity periods may be available by arrangement with the developer.

To activate the software, open the program and, when the "License Required" window appears, enter the supplied key in the relevant field and click "Activate." If the key is valid, a confirmation message displays the activation date, expiry date, and remaining number of days. The software can then be used without feature restrictions for the licensed period. Each key can be used only once; after use, it is recorded in the local software database to prevent reuse. A new key is therefore required when the current license expires.

License details and current status can be viewed through Help → License Info in the application menu. This window shows whether the software is in its trial period or has an active or expired license, along with the expiry date and remaining days. For activation problems or license renewal, contact the developer at the email address above.

8. Summary and Quick-Use Guidance

SCS Flood Hydrograph estimates flood hydrographs using either the lumped or distributed method. For the lumped method, load the rainfall file containing time and cumulative rainfall, then enter CN and watershed area directly or load a CN raster so that these values can be calculated automatically. For the distributed method, the CN and flowlength rasters are required. The output consists of a discharge–time table and graph, and the results can be saved.

Important recommendation: for the distributed method, ensure that the flow-length raster, CN raster, and—if available—the distributed rainfall raster have the same coordinate reference system, extent, and cell size. Set the storage coefficient R according to watershed conditions.

9. References

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